Virtual lokal tarmoq

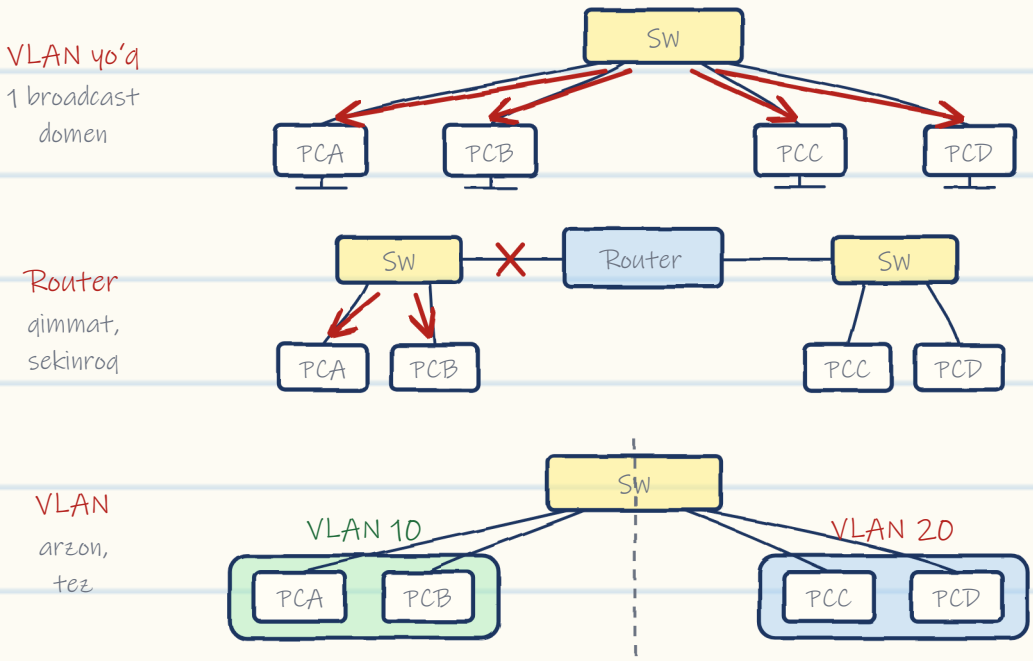
Virtual Local Area Network

Asosiy atamalar

- VLAN — bitta fizik tarmoq ichidagi mantiqiy alohida guruh
- Teg (tag) — kadrga qo'shiladigan 4 baytli 802.1Q belgisi
- VLAN ID — tegdagi 12 bitli raqam (1–4094)
- access · trunk — tegsiz bitta VLAN · tegli bir necha VLAN
- PVID — portning o'z VLAN raqami, tegsiz kadrga qo'yiladi

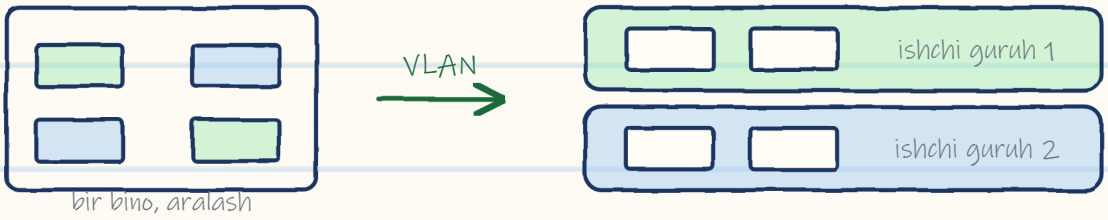
1. Nega VLAN kerak?

Kommutator kolliziya domenini bo'ladi, broadcast domenini bo'lmaydi.

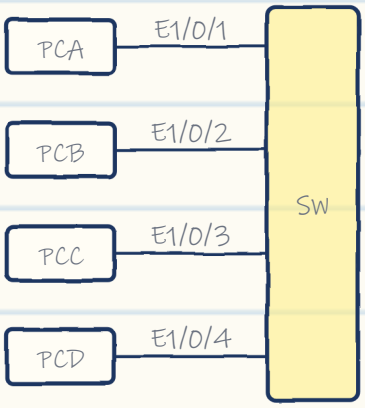


1-rasm. Broadcastni chegaralashning uchta yo'li

- 400 PC × 100 kbit/s broadcast = 40 Mbit/s (100 Mbit/s liniyaning 40 %)
- VLAN = alohida 2-qatlam broadcast domeni = ko'prik domeni
- VLAN'lar orasida — faqat router yoki 3-qatlam kommutator



2. VLAN taqsimlash usullari



port asosida

E1/0/1–2	VLAN 10
E1/0/3–4	VLAN 20

protokol asosida

IP	VLAN 10
IPX	VLAN 20

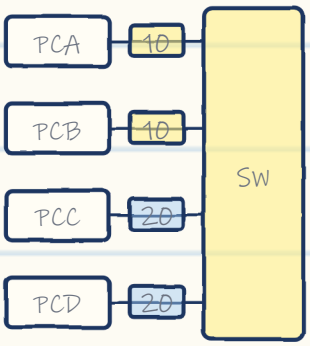
IP tarmoq asosida

10.0.0.0/24	VLAN 10
20.0.0.0/24	VLAN 20

port asosidagi VLAN — qolgan usullar uchun asos

- Port — portga VLAN beriladi; eng ko'p ishlatiladi
- Protokol — kadr ichidagi protokol (IP, IPX) bo'yicha
- IP tarmoq — manba IP va niqob bo'yicha

3. VLAN tag va MAC jadval



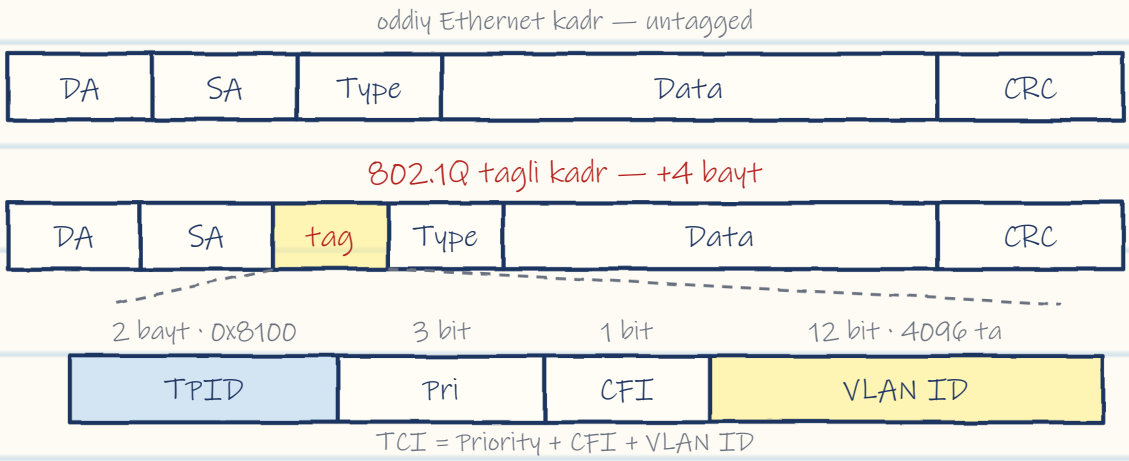
MAC	VLAN	Port
MAC-A	10	E1/0/1
MAC-B	10	E1/0/2
MAC-C	20	E1/0/3
MAC-D	20	E1/0/4

qidiruv = maqsad MAC + VLAN ID

2-rasm. Kirish portida tag qo'yiladi, jadvalga VLAN ID bilan yoziladi

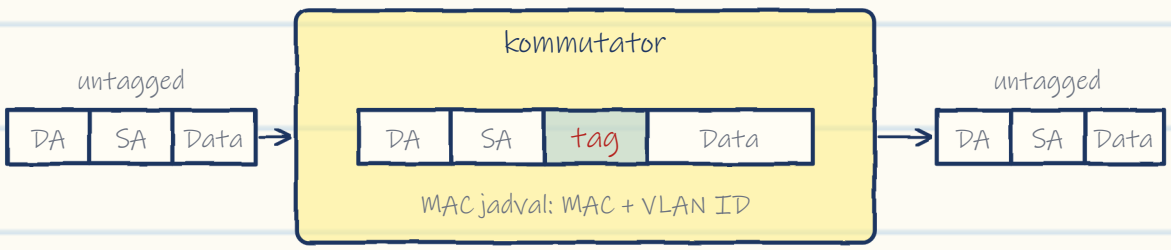
Eslatma: bitta VLAN bir necha kommutatordan o'tishi mumkin — kommutatorlar orasida tag saqlanadi.

4. IEEE 802.1Q kadr formati



- TPID 2 bayt — 0x8100 bo'lsa kadr tagli
- Priority 3 bit — 802.1p, qiymat 0...7
- CFI 1 bit — 0 standart MAC formati, 1 nostandart
- VLAN ID 12 bit — 0...4095, jami 4096 ta

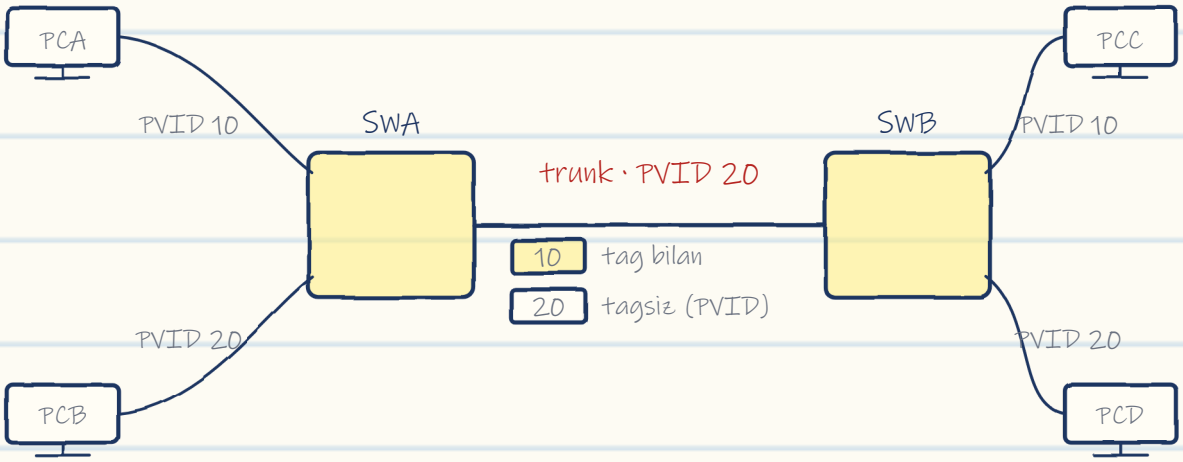
5. Tag bilan ishlash



3-rasm. Xost tagni ko'rmaydi — hammasi kommutator ichida

Standart: kommutatorda faqat VLAN 1 bor, hamma port — access, PVID = 1.

6. Port turlari: PVID bilan o‘yin



access — faqat PVID · trunk — PVID tagsiz, qolgani tagli

4-rasm. PCA–PCC VLAN 10 da, PCB–PCD VLAN 20 da gaplashadi

7. access · trunk · hybrid



5-rasm. Farq faqat tagsiz chiqadigan VLAN’larda

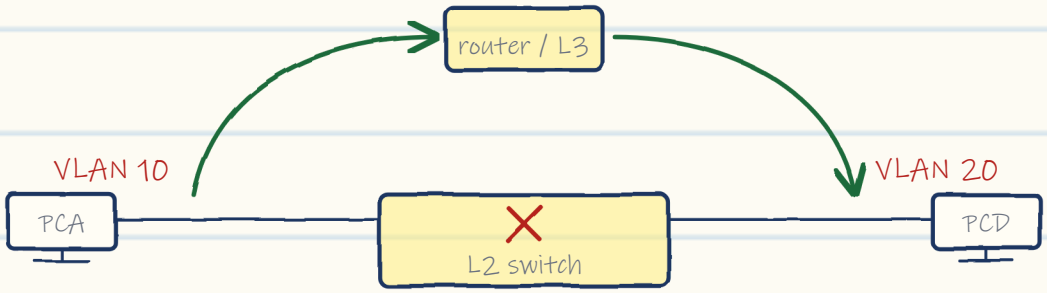
Hybrid: turli VLAN’dagi xostlar bitta umumiy server bilan gaplashadi. Trunk va hybrid orasida — avval access.

	access	trunk	hybrid
Tagsiz chiqadi	PVID	faqat PVID	xohlagan VLAN

8. Sozlash

- vlan 10 → port Ethernet1/0/1
- port link-type access | trunk | hybrid
- port trunk permit vlan 10 20 · port trunk pvid vlan 20
- port hybrid vlan 10 20 untagged · port hybrid pvid vlan 30
- display vlan · display interface Ethernet1/0/1

9. VLAN'lar orasida aloqa



2-qatlamda bloklangan — 3-qatlam orqali o'tadi

Xulosa

- VLAN = alohida broadcast domen, kommutatorda arzon
- Tag 4 bayt: TPID 0x8100 + Priority 3 bit + CFI 1 bit + VLAN ID 12 bit
- MAC jadval qidiruvi — MAC va VLAN ID juftligi bo'yicha
- access 1 VLAN · trunk PVID tagsiz · hybrid xohlagani tagsiz

Tekshirish

1. TPID qiymati? — 0x8100
2. VLAN ID necha bit, nechta VLAN? — 12 bit, 4096
3. Trunk qaysi VLAN'ni tagsiz yuboradi? — PVID
4. Trunk'ni hybrid qilish uchun? — avval access qilinadi